

OpenFlex Wheeled Humanoid Robot User Manual (Advanced)

English | [中文](#)

This document covers the perception, mapping, localization, navigation, VLA data collection/training/inference, and core algorithm details for the OpenFlex robot system.

For hardware control, VR teleoperation, and subsystem basics, refer to the Basic manual.

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Chapter 1: Sensor System

1.1 Quick Start

The primary sensor is the Livox Mid-360 3D LiDAR with built-in IMU. Network configuration is required before starting the driver.

Item	Value
Host IP	192.168.1.50
Netmask	255.255.255.0
MID360 IP	192.168.1.173

```
sudo ip addr add 192.168.1.50/24 dev eno1
sudo ip link set eno1 up
ping 192.168.1.173

ros2 launch livox_ros_driver2 msg_MID360_launch.py
ros2 topic hz /livox/lidar      # ~10 Hz
ros2 topic hz /livox/imu       # ~200 Hz
```

1.2 Livox Mid-360 Specifications

Parameter	Value
Range	0.3~40 m
FOV	360° × 59°
Point rate	~200,000 pts/s
Built-in IMU	6-axis, 200 Hz
Interface	Ethernet
ROS message	livox_ros_driver2/msg/CustomMsg

Config:

src/openflex_chassis/hardware_sensor_layer/livox_ros_driver2/config/MID360_config.json

1.3 Other Sensors

- Intel D435: optional (enable via use_camera:=true)
- RPLiDAR C1: backup 2D LiDAR (URDF placeholder)

Chapter 2: Mapping

2.1 Quick Start

```
export MAP_NAME=your_map
mkdir -p "$HOME/MID_360_nav/openflex_maps/3d/$MAP_NAME"
ros2 launch swerve_navigation mapping.launch.py

# Save map
ros2 service call /pgo/save_maps interface/srv/SaveMaps \
  "{file_path: '$HOME/MID_360_nav/openflex_maps/3d/$MAP_NAME',
  save_patches: true}"

# Convert 3D→2D
mkdir -p "$HOME/MID_360_nav/openflex_maps/2d/$MAP_NAME"
ros2 run pgo pcd_to_nav2_map \
  --pcd "$HOME/MID_360_nav/openflex_maps/3d/$MAP_NAME/map.pcd" \
  --out-dir "$HOME/MID_360_nav/openflex_maps/2d/$MAP_NAME" \
  --resolution 0.05 --obstacle-z-min 0.10 --obstacle-z-max 1.40
```

2.2 FAST-LIO2 Algorithm

Tightly-coupled LiDAR-Inertial Odometry using IESKF + ikd-Tree.

Code: src/openflex_chassis/mapping_localization_layer/fastlio2/src/lio_node.cpp

Per-frame pipeline: IMU pre-integration → point undistortion → IESKF update (point-to-plane) → ikd-Tree map update.

Parameter	Mapping	Navigation
world_frame	map	odom
body_frame	mid360_link	base_link
ieskf_max_iter	4	3
publish_map_odom	true	false

2.3 PGO

Loop closure via ScanContext + ICP. Publishes `/pgo/offset` correction. Key params:
`loop_search_radius=1.0m`, `loop_score_tresh=0.15`.

Code: `src/openflex_chassis/mapping_localization_layer/pgo/`

2.4 3D → 2D Map Conversion

Z-slice projection → SOR filtering → raycasting → inflation. Output: `map.yaml` + `map.pgm` for Nav2.

Chapter 3: Localization

3.1 Architecture

```
ICP: live cloud + PCD map → map→odom
safe_odom_mux: LIO + wheel odom → odom→base_link (50 Hz)
Nav2 uses: map→base_link = map→odom × odom→base_link
```

3.2 ICP Three-Level Strategy

Code: `src/openflex_chassis/mapping_localization_layer/icp_registration/`

- 1. ScanContext → NDT → ICP (~3-5s)
- 2. ScanContext → ICP directly (NDT fails)
- 3. Grid search ICP (~10-15s, last resort)

Key params: `thresh=0.5`, `xy_offset=5.0m`, `yaw_resolution=30°`,
`continuous_realign_interval=5.0s`.

3.3 Continuous Realignment

Every 5s: skip ScanContext, use current map → odom as initial guess, run ICP refinement. Non-blocking (separate thread). Failure preserves last valid transform.

3.4 Navigation-Mode TF Tree

Transform	Provider	Rate
map → odom	icp_registration	~100 Hz

Transform	Provider	Rate
odom → base_link	safe_odom_mux	50 Hz
base_link → sensors	robot_state_publisher	static

Chapter 4: Navigation

4.1 Quick Start

```
ros2 launch swerve_navigation full_system.launch.py \
  pcd_path:=/path/to/map.pcd \
  map_yaml:=/path/to/map.yaml \
  initial_pose="[0,0,0,0,0,0]"
```

4.2 Nav2 Stack

Config:
`src/openflex_chassis/navigation_layer/swerve_navigation/config/nav2_params.yaml`

Component	Plugin
Controller	<code>nmpc_controller::NmpcController</code>
Planner	<code>nav2_smac_planner/SmacPlanner2D</code>
Smoother	SimpleSmoother
Behaviors	Spin, Backup, Wait
Velocity Smoother	VelocitySmoother

4.3 NMPC Controller

Code: `src/openflex_chassis/navigation_layer/nmpc_controller/`
Model: `state=[px,py,θ,vx,vy,ω], control=[ax,ay,α]`. acados SQP_RTI, N=20, Tf=2.0s.

Limit	Value
<code>vx_max</code>	0.50 m/s
<code>vy_max</code>	0.25 m/s
<code>omega_max</code>	1.0 rad/s

Features: costmap cost penalty, local A* obstacle repair, rotate-to-heading, swerve-native lateral tracking.

4.4 Costmap

- Local: 5×5m rolling, `/scan` obstacle layer, inflation 0.30m

- Global: static layer + obstacle + inflation
- Footprint: 0.74×0.62m rectangle

4.5 Velocity Smoother

`max_velocity=[0.50, 0.22, 1.0], max_accel=[0.8, 0.6, 1.5]`, open-loop feedback.

Chapter 5: Whole-Body VLA Data Collection, Training and Inference

5.1 Overview

Uses LeRobot's Robot/Follower + Teleoperator/Leader abstraction. VR teleoperation provides real-time control; LeRobot records observations and actions. During inference, trained policy outputs actions back to ROS2.

Note: VLA data collection, training, and inference are optional extension modules. The current OpenFlex base source package does not include the `openflex_vla` implementation. Commands using `<openflex_vla_package>` must be replaced with the actual ROS package name provided by the installed VLA extension.

5.2 Quick Start

```
# Hardware + SLAM + cameras
ros2 launch <openflex_vla_package> wholebody_vla_record.launch.py

# VR teleoperation
ros2 launch openarmx_integrated_bringup integrated_vr_teleop.launch.py

# Record
python3 scripts/lerobot_record_openflex.py \
  --robot.type=openarmx_follower_ros2 \
  --robot.skip_send_action=true \
  --robot.ros2.enable_base=true \
  --robot.ros2.base_action_mode=pose \
  --dataset.repo_id=YOUR_NAME/task --dataset.fps=15
```

5.3 Schema

Observation: 16 arm joints + 2 head + 2 lift + 3 base twist + 3 episode-local pose + 4 RGB cameras = 26 scalars + 4 images.

Action (common): 16 arm + 2 head + 1 lift = 19 dims. Base action depends on mode:

- `pose`: +3 (base_target_x/y/theta) = 22 total
- `trajectory`: +24 (8 future points × 3) = 43 total
- `twist`: +3 (vx/vy/wz) = 22 total

5.4 Training & Inference

Policies: ACT (50-200 eps), Diffusion Policy (100-500), VLA models (500+).

Inference: close VR, set `skip_send_action=false`. `pose/trajectory` modes publish `/vla/base_path`; `twist` publishes `/cmd_vel` directly.

Appendix: Core Algorithm Details and References

A.1 Swerve Kinematics

IK: $v_{x_i} = v_x - \omega * y_i$, $v_{y_i} = v_y + \omega * x_i$, $angle = \text{atan2}(v_{y_i}, v_{x_i})$, $speed = \text{sqrt}(\dots)$.

FK: Least-squares pseudo-inverse. Midpoint integration for odometry.

Refs: Muir & Neuman 1987; Campion et al. 1996.

A.2 FAST-LIO2

IESKF state: position, velocity, rotation, IMU biases, LiDAR-IMU extrinsics. Point-to-plane residuals in ikd-Tree.

Refs: Xu et al. "FAST-LIO2" T-RO 2022; Cai et al. "ikd-Tree" 2021.

A.3 FAST-LIVO2

LiDAR + IMU + camera fusion. Visual constraints complement geometric features in degraded environments.

Refs: Zheng et al. "FAST-LIVO/FAST-LIVO2" 2022/2024.

A.4 ScanContext + NDT + ICP

ScanContext: polar descriptor (20 rings $\times$ 60 sectors), RingKey KD-tree search. NDT: normal distribution grid matching. ICP: iterative closest point with point-to-plane.

Refs: Kim & Kim 2018/2021; Biber & Strasser 2003; Besl & McKay 1992.

A.5 PGO / iSAM2

Pose graph with odometry + loop factors. Incremental Bayes tree optimization.

Refs: Kaess et al. "iSAM2" IJRR 2012; Dellaert "GTSAM" 2012.

A.6 NMPC / acados

Nonlinear MPC, SQP_RTI single iteration per cycle, HPIPM QP solver. Costmap-aware stage cost.

Refs: Mayne et al. Automatica 2000; Verschueren et al. "acados" MPC 2022.

A.7 VR IK & Safety

Pinocchio damped least-squares IK, step limiting, joint clamping. Head: relative quaternion control with soft limits.

Refs: Carpentier et al. "Pinocchio" 2019; Siciliano et al. 2009.

A.8 LeRobot VLA

ACT action chunking, Diffusion Policy denoising, episode-local odom design avoiding global drift.

Refs: Zhao et al. "ACT" RSS 2023; Chi et al. "Diffusion Policy" RSS 2023; Cadene et al. "LeRobot" 2024.

A.9 Full Reference List

Domain	References
Swerve kinematics	Muir & Neuman 1987; Campion et al. 1996
LiDAR-Inertial	Xu et al. FAST-LIO2 2022; Cai et al. ikd-Tree 2021
Visual-Inertial	Zheng et al. FAST-LIVO 2022; FAST-LIVO2 2024
Place recognition	Kim & Kim ScanContext 2018/2021
Registration	Besl & McKay ICP 1992; Biber & Strasser NDT 2003
Graph SLAM	Kaess et al. iSAM2 2012; Dellaert GTSAM 2012
Navigation	Hart et al. A* 1968; Macenski et al. Nav2 2020
NMPC	Mayne et al. 2000; Verschueren et al. acados 2022
Robotics/IK	Carpentier et al. Pinocchio 2019; Nakamura & Hanafusa 1986
CANopen	CiA 301; CiA 402
VLA	ACT 2023; Diffusion Policy 2023; RT-1/RT-2 2023; OpenVLA 2024; pi0 2024; LeRobot 2024

End of Advanced Manual.